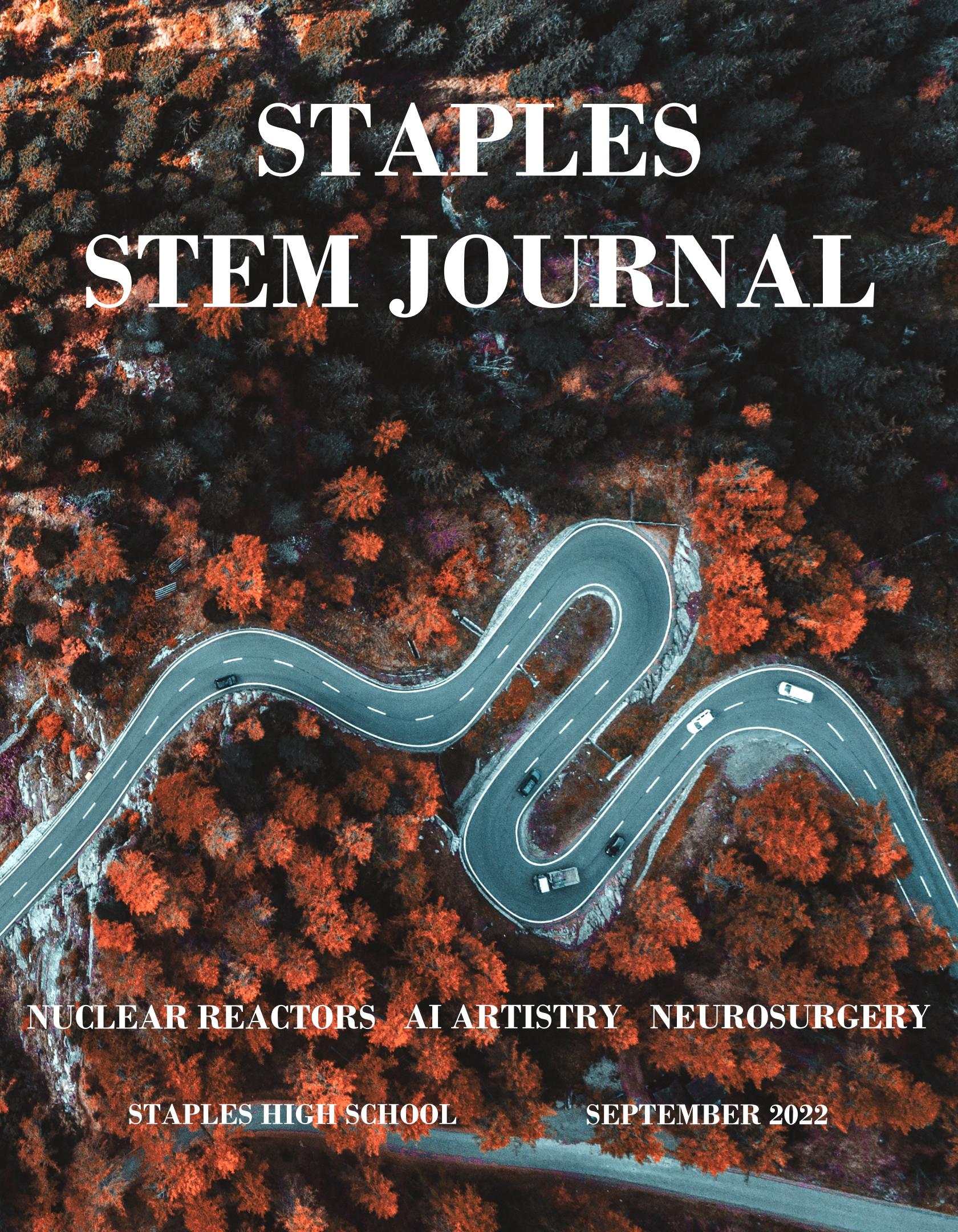
STAPLES STEM JOURNAL



NUCLEAR REACTORS AI ARTISTRY NEUROSURGERY

STAPLES HIGH SCHOOL SEPTEMBER 2022

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The Staples STEM Journal provides an outlet for individuals to share their STEM interests with the Staples community and aims to broaden public interest and knowledge in these fields.

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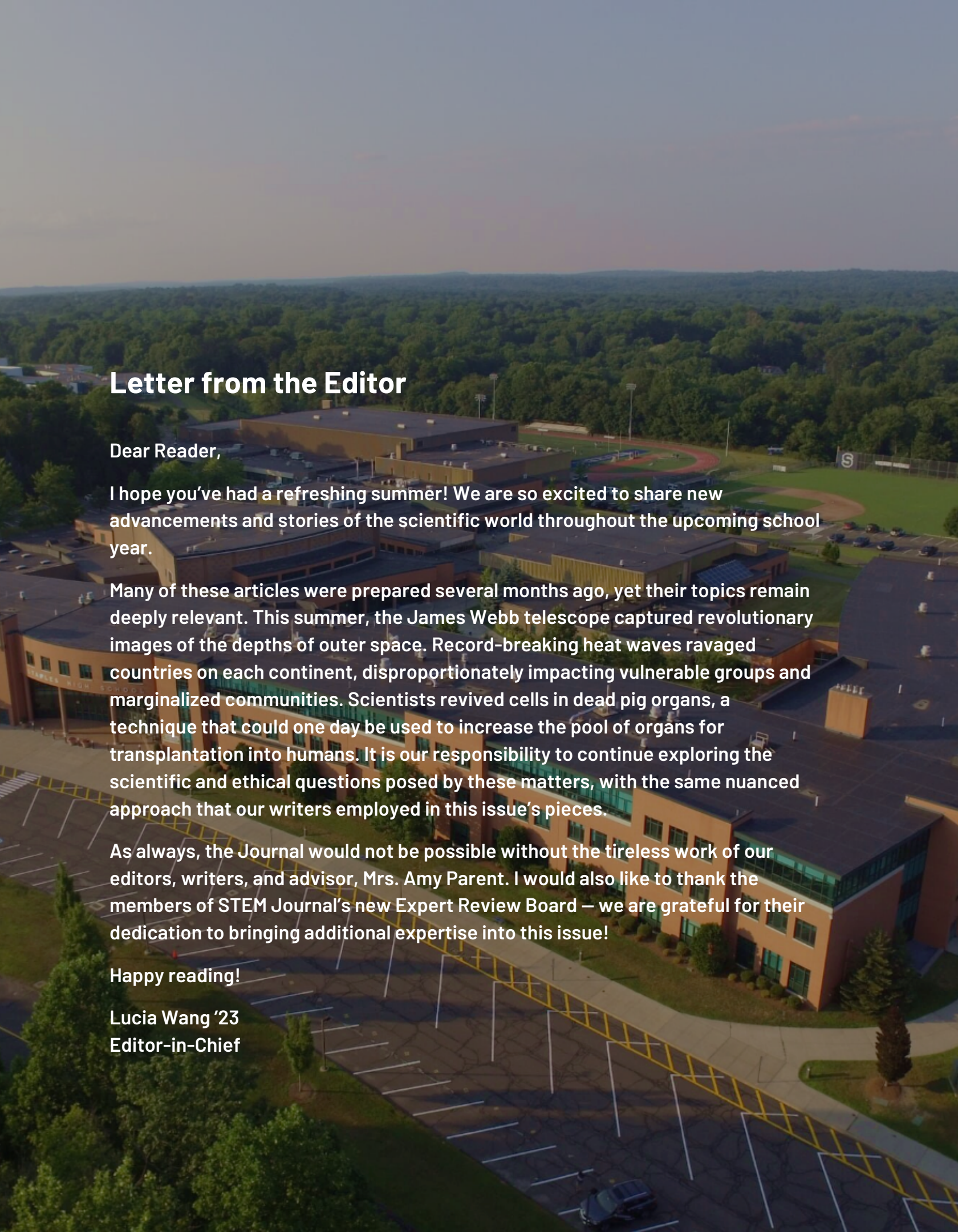

Letter from the Editor

Dear Reader,

I hope you've had a refreshing summer! We are so excited to share new advancements and stories of the scientific world throughout the upcoming school year.

Many of these articles were prepared several months ago, yet their topics remain deeply relevant. This summer, the James Webb telescope captured revolutionary images of the depths of outer space. Record-breaking heat waves ravaged countries on each continent, disproportionately impacting vulnerable groups and marginalized communities. Scientists revived cells in dead pig organs, a technique that could one day be used to increase the pool of organs for transplantation into humans. It is our responsibility to continue exploring the scientific and ethical questions posed by these matters, with the same nuanced approach that our writers employed in this issue's pieces.

As always, the Journal would not be possible without the tireless work of our editors, writers, and advisor, Mrs. Amy Parent. I would also like to thank the members of STEM Journal's new Expert Review Board — we are grateful for their dedication to bringing additional expertise into this issue!

Happy reading!

Lucia Wang '23
Editor-in-Chief

1. Sky Oddity

Colin Morgeson, ‘23

A vast shadow passes over the marshes of prehistoric Texas, and grazing dinosaurs turn their attention toward the skies. Gliding over the land is the Quetzalcoatlus, the largest known flying animal to ever exist. With a 40 foot wingspan and a massive beak, the Quetzalcoatlus resembled a distorted, gargantuan heron as it roamed the skies approximately 70 million years ago [1].

Today, the Quetzalcoatlus no longer soars overhead but continues to captivate the minds and imaginations of both paleontologists and the general public. First discovered in 1971, the sizable pterosaur (not pterodactyl, as despite common misconceptions, such a classification does not actually exist) has historically enjoyed surprisingly little scientific investigation. [3] According to Kevin Padian, a biology professor and emeritus curator at the University of California’s Museum of Paleontology, “[The Quetzalcoatlus] is legendary, although most of the public conception of the animal is artistic, not scientific.” Indeed, many pertinent questions about such an oddity of an animal have long gone unanswered. Perhaps most notably, the matter of how the Quetzalcoatlus managed to get itself airborne might seem simple but has long been a contentious subject among paleontologists. [1] Fortunately, a new collection of papers published by the Society of Vertebrate Paleontology in December of 2021 have managed to shed light on the lesser known aspects of the Quetzalcoatlus. In this study, the researchers examined fossils of Quetzalcoatlus northropi and Quetzalcoatlus lawsoni, two similarly built yet different sized species of Quetzalcoatlus. (Q. lawsoni measures about half as large as Q. northropi.) [6] Padian, who worked on the study, claims “[it] is the first real look at the entirety of the largest animal ever to fly, as far as we know” [5].

Feeding

Paleontologists have several theories about what and how pterosaurs ate, which vary by species. In order to determine the Quetzalcoatlus’s diet, the researchers examined its thin beak, which they noted was distinctly “chopstick”-like, limiting the size of its prey. [1] The researchers also took the Quetzalcoatlus’s habitat into account, which consisted of inland marshes and open fields. Given these facts, the team concluded that the Quetzalcoatlus most likely hunted as a wader and a stalker, consuming prey such as small fish, insects, and

A realistic depiction of the Quetzalcoatlus [5]

lizards, similarly to a modern day heron or egret. [3] According to Padian, "Some people said it was a carrion feeder, some people said it flew over the water and plucked fish like a pelican. Those things don't work. The jaws are very long and thin, tapering to a point…If you look at a heron or egret's jaws, they're the same — good for plucking lizards and other small game, but definitely not carcass-scavenging. It had no teeth." However, the researchers acknowledge other possible methods of hunting besides wading and stalking; because of the impressive mobility of the Quetzalcoatlus's neck and head, it may have used a more dynamic method of hunting. "It could lower the great head far below the horizontal, so if it were cruising above dry land, it might have been able to swoop down and pluck an unsuspecting animal," Padian stated. [5]

Launching

The ability to launch off the ground and take flight is fundamental for any airborne animal, but for the Quetzalcoatlus, this task wasn't simple. The Quetzalcoatlus could not simply run along the ground whilst flapping its wings to take flight, as modern-day birds do. Although the researchers concluded that the Quetzalcoatlus could probably run, it faced a difficulty due to its short legs compared to its massive wings; simply flapping while aground would cause its wings to collide with the ground beneath it. (In fact, this complication has led many paleontologists to theorize that the Quetzalcoatlus couldn't fly at all, in spite of its prominent wings.) [4] Thus, getting airborne required a unique approach. One possible approach could be that the Quetzalcoatlus used its forearms to push off the ground. However, the researchers have deemed this unlikely, as it would require the pterosaur to have extremely strong forelimbs. Instead, they theorize that it first assumed a crouch, then pushed off the ground with its muscular hind legs in a sizable jump, gaining enough height for its wings to clear the ground and begin broad, powerful flaps. It is essential that the jump carried the pterosaur at least 8 feet off the ground, and Padian noted that this method "depends on sufficient power from the legs." Once again, this behavior is similar to that of modern day herons and egrets [5].

A skeletal reconstruction of the Quetzalcoatlus [5]

Walking

For an animal as oddly proportioned as the Quetzalcoatlus, the simple act of walking wasn't simple at all. Many winged animals, such as the vampire bat, use their forelimbs to help propel them forward while walking, but the Quetzalcoatlus's unique bone structure prevented this. However, its forelimbs did have to come into contact with the ground, as they were simply too large for holding them up to be practical. Because of this, the research team theorizes that its forelimbs acted similarly to ski poles, simply supporting it as it walked. [5] As described by Padian, "To avoid tripping, the animal first raised its left arm, then advanced its left leg in a full step, then it placed the hand on the ground. The process was repeated with the right limb: The right arm lifted, the right leg advanced and emplaced the right foot, and then the right hand descended." This process might seem overly complicated, but the Quetzalcoatlus could perform the gait with ease. [2] Additionally, this theory also aligns with oddly shaped fossilized tracks left by pterosaurs in France,

lending further credence to it. While other behaviors of the Quetzalcoatlus can be observed in modern day birds, its suspected walking pattern is unlike any species that exists today. [6]

While the Quetzalcoatlus itself was discovered in 1971, the findings of the Society of Vertebrate Paleontology's study act as something of a rediscovery, as they have provided novel illustrations into the life of such a distinctive creature. People have unearthed and studied dinosaur and pterosaur remnants for centuries, but even today, there are new discoveries to be made. As Padian put it, "The results [of our research] are revolutionary for the study of pterosaurs — the first animals, after insects, ever to evolve powered flight." Even today, scientists remarkably continue to flesh out the existences of animals lost to time. [1]

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2. The NeuroPace RNS System: Surgery for Depression

Samantha Sandrew, ‘25

Thirty percent of people with depression don’t respond to medication or psychotherapy. Electroconvulsive Therapy is an option but comes with potential side effects such as memory loss [1]. The NeuroPace RNS System, a brain implant used to treat epilepsy and Parkinson’s disease, is being tested as a viable therapy for treatment resistant clinical depression. The application of the NeuroPace RNS system is innovative but has an uncertain future.

An ongoing clinical trial, led by scientists at the University of California at San Francisco, called Closed-Loop Deep Brain Stimulation for Major Depression, or PRESIDIO, is exploring a new treatment of Major Depressive Disorder (MDD) [1]. This active study is enrolling people who have treatment resistant depression. It could create hope for those who have exhausted all possibilities.

Premise of the Trial

During the study, biomarkers, which are ‘flags’ that alert the occurrence of biological processes, were used to find areas of the brain where the patient's symptoms are the most intense. These symptoms, often lack of energy, irregular habits, and negative feelings, typically originate in the amygdala, orbitofrontal cortex, and hippocampus [2]. The researchers utilized an open loop technique at first. Open loop therapy is a system that follows the commands it is told; it has no capacity to monitor or regulate itself through feedback. This technique is used when the researchers stimulate different parts of the brain and receive patient feedback. They found that the ventral capsule/ventral striatum, or VC/VS, was the area that suppressed the patient's symptoms the most [2]. The continued stimulation throughout the study resulted in the patient's depressive state to be temporarily relieved. These promising results prompted a turn into closed loop therapy. Closed loop therapy is the opposite of open loop, as it adjusts to the feedback it receives. Closed loop provides stimulation only when necessary; it reacts to the environment, whereas open loop therapy has no self-correcting action and is manually adjusted. The best fit depends on the aimed result. In this trial, the closed loop therapy took form as a surgical implant which stimulates the targeted brain areas.

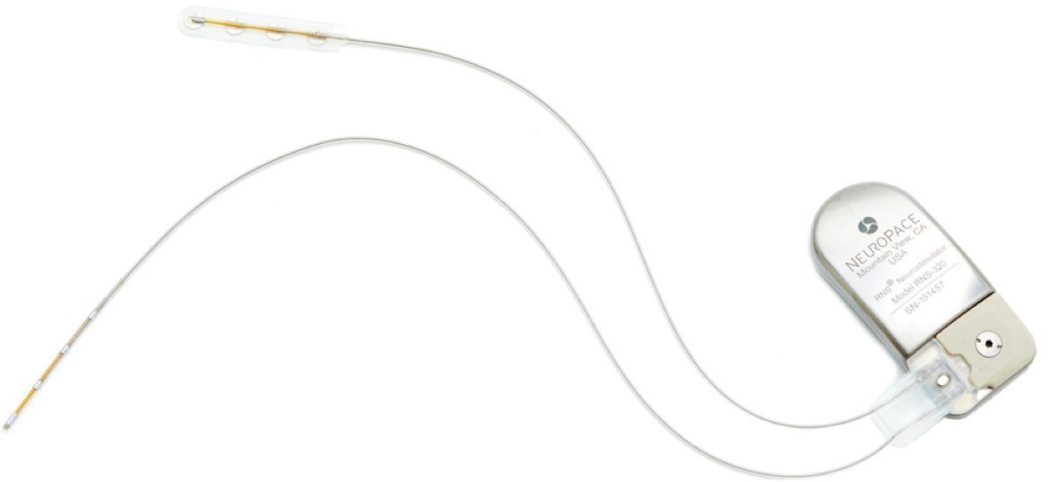
The Patient's Journey

Sarah checking in with Kathrine Scangos at the Langley Porter's psychiatric hospital [4].

In June of 2020, Sarah became the first person to undergo this experimental therapy. Sarah was diagnosed with severe onset MDD in early childhood [3]. All possible options, including combinations of antidepressants and electroconvulsive therapy, failed to aid Sarah. She enrolled in the UCSF trial in 2019 [1] (Figure 1). Her life changed when they shocked her VC. She said, "As they kept playing with that area, I gradually looked down at the needlework I had been doing as a way to keep my mind off negative thoughts and realized I enjoyed doing it, which was a feeling I haven't felt for years" [1]. Then, Sarah had surgery to implant the NeuroPace RNS system. Before surgery, she marked 36 out of 54 on the Montgomery–Åsberg Depression Rating Scale, a questionnaire used to rank the severity of depression. Twelve days after the surgery, the number decreased to 14. Months later, she scored just below 10, which is considered a full remission [3]. It has now been a year since Sarah had the surgery and the device is in full effect. "As time has gone on, it's been this virtuous cycle, a spiral upwards," Sarah said. "Everything has gotten easier and easier and easier" [5].

What makes this trial special?

Stimulating the brain to treat depression is not new; it's called deep brain stimulation (DBS). The NeuroPace RNS is a form of DBS. Other forms of DBS have been tested against depression; however, they were inconclusive. The researchers think the difference is found when and where stimulation occurs. The Neuropace RNS system works through one wire detecting activity in the region that negative systems stem from; the other wire stimulates the patient's reward section of the brain (Figure 2). Dr. Katherine Scangos explained, "We think that a reason for this is that traditional DBS delivers continuous stimulation, 24 hours a day, rather than kicking in when a patient's symptoms worsen and the location of the stimulation is not personalized to each individual's depression" [2].



The Neuropace RNS system [6].

Another special quality the NeuroPace RNS system displays is its ability to reteach behavior. Through keeping the neural circuit functioning correctly, "pathways that support pathological negative thought processes in depression can be unlearned" [1]. The Neuropace RNS system can correct the patient's way of thinking, easing the intensity of the patient's depression. Medicines and other treatments require commitment and are not always an option. The unorthodox personalization, time aspect, and ability to reteach behavior aspects of the treatment is inventive, allowing the researchers to take an effective approach to treating MDD.

Challenges

While the personalization of the technology may be the reason for its success, it could also be an issue in the future. Finding biomarkers in each individual takes sophisticated imagery and machine learning technology. Helen Mayberg, a neurologist at Icahn School of Medicine, notes that the complexity may prevent it from being more widely utilized. She said, "Time will tell if this approach substantially improves on simpler open-loop strategies [without biomarkers]" [7].

The clinical trials have just begun and the scientists are still observing the technology's effect. However, researchers are hopeful for the future. "In line with our model, this patient's depression acutely improved once closed-loop therapy was initiated," the research paper concluded. "The effect has been maintained with sustained closed-loop therapy through the time of writing this article" [3]. The researchers at the PRESIDIO trial are working to maintain the success it has had in the past year, providing light for those whose days are darkened by depression.

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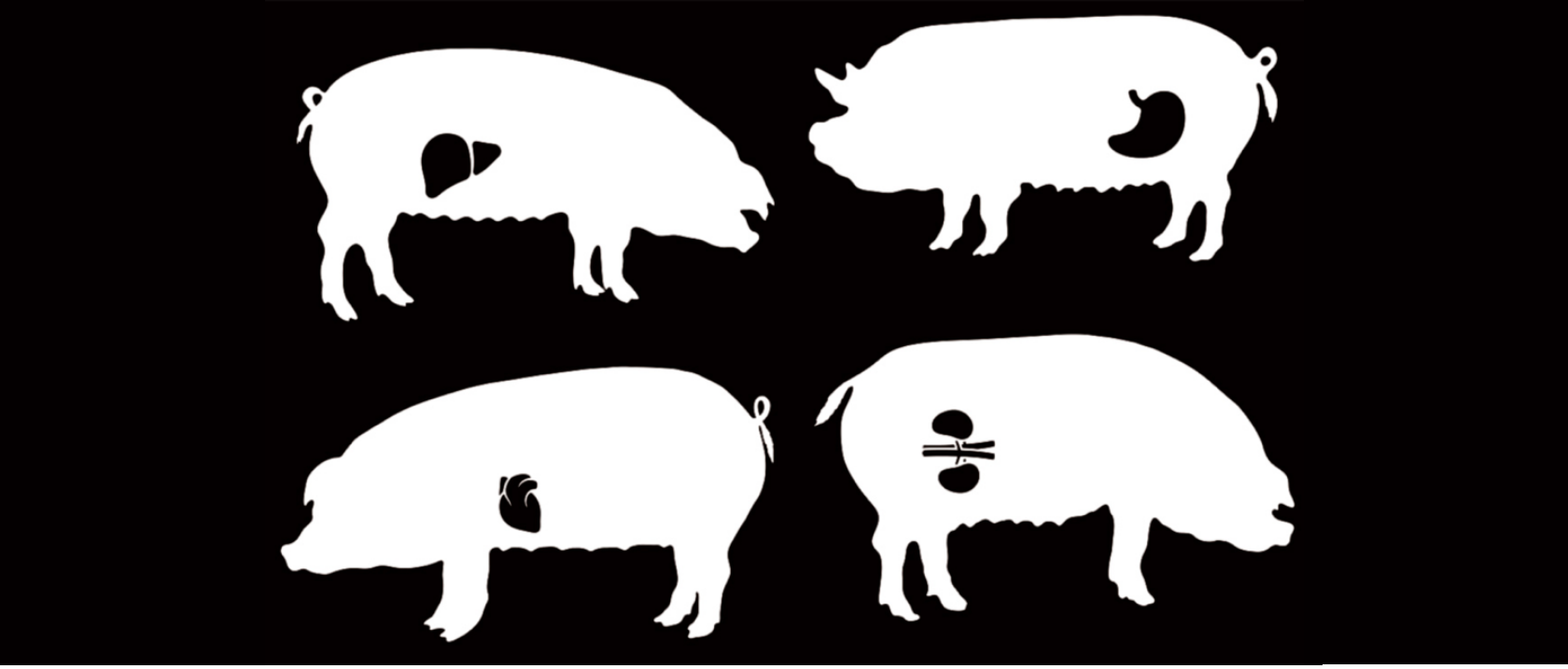


3. Porcine Xenotransplantation: History, Barriers, and Recent Progress

Samuel Zwick-Lavinsky, ‘25

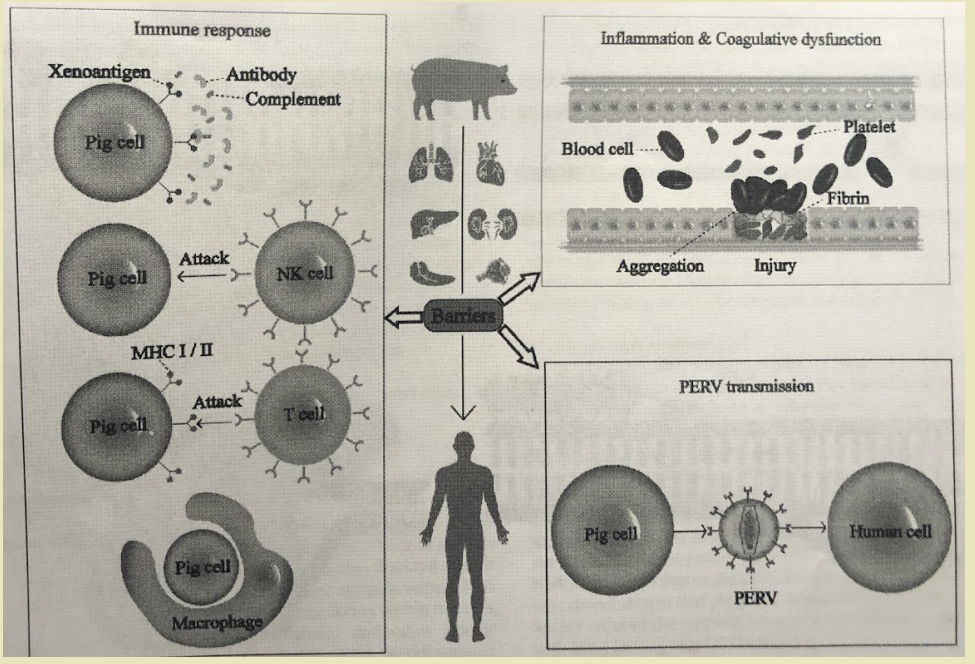
Introduction

Humanity’s fascination with transferring bodily organs from animals to humans extends back to antiquity. As one scholar has noted, this fantasy is demonstrated in the myth of Daedalus and Icarus [1]. Throughout history, there have been countless stories of hybrid humans and animals, such as the ancient Egyptian god Anubis, the Hindu deity Ganesha, and the Greek sphinx. There is also an interest in those who turn into animals after receiving some part of them, such as in Mary Shelley’s Frankenstein or 1986 film “The Fly.” While xenotransplantation, or the transplantation of living cells, tissues or organs from one species to another, has long been viewed as science fiction, scientists recently have reached a series of medical breakthroughs that offer a promising strategy to solve the massive organ shortage problem. At the University of Maryland in early January, a 57-year-old man with life-threatening cardiac disease received a beating heart from a genetically modified pig. And later that month, scientists at the University of Alabama performed the first clinical-grade porcine kidney xenotransplant into a brain-dead patient with no chance of revival.

For the 700,000 Americans who live with kidney failure, kidney transplantations are extremely elusive and most patients depend on dialysis, a costly and inconvenient treatment [2]. Currently, only 25,000 kidney transplants are done in the United States per year and more than a dozen people on the transplant waiting list die each day [3]. This dire issue is closely wrapped up with social justice and medical ethics-- because of issues like diabetes and high blood pressure, kidney disease disproportionately affects people of color, women, and those with less education and lower incomes [3]. As such, scientists have been attempting clinical trials using non-human primate models (NHPs) for the past century. While one might assume that NHPs would be the best source of organs for xenotransplantation, recent research has revealed that the domestic pig is in fact an ideal organ source because of its physiological similarity to humans and its high reproductive capacity. In addition, pigs reach sexual and physical maturity in only about 4-8 months; therefore, an unlimited availability of organs for xenotransplation is feasible in a relatively short time period [4].

The hurdles in pig-to-human xenotransplantation

While pigs have similar sized organs to humans, the two species are completely different genetically. This results in incompatibilities between them that must be suppressed or reversed in order to transplant organs from pigs to humans. The first difficulty is immune rejection, which occurs when the immune system recognizes a transplanted organ as foreign and launches an attack on it. This response is hence called antibody-mediated rejection (AMR), the main types of which are hyperacute rejection (HAR), acute rejection, and chronic rejection. Hyperacute rejection is the most dangerous and most immediate type of rejection: it prompts an immediate decline in the function of the organ and eventually organ necrosis, which can be quite problematic for the transplant recipient. As opposed to HAR, acute rejection happens weeks or months after organ transplantation-- it is often manifested as fever, swelling, and dysfunction of the organ. Chronic rejection occurs over time months to years after the transplant, but can be the most dangerous because it is irreversible at this point. [4]. All of these reactions must be avoided at all costs. While immune rejection can occur in allotransplantation, or the transplant of organs from a donor to a recipient of the same species, the immune response during xenotransplantation is much stronger because of the molecular differences across species as well as the abundance of foreign antigens in non-human organs. Specifically, the binding of preformed antibodies to the pig antigen galactose-*α*-1,3, galactose (Gal) is most responsible for HAR [4]. Furthermore, most graft failures are associated with blood coagulation or inflammation in the blood vessels, a critical barrier to xenotransplantation. Finally, cross-species transmission of zoonotic diseases, especially porcine endogenous retroviruses (PERVs), is a significant risk that is difficult to manage and must be treated with extreme caution. Because immunosuppressants are typically used in xenotransplantation, PERVs become very dangerous and even deadly [4].



The barriers in porcine xenotransplantation [4].

Xenotransplantation's history

Most early xenotransplantation attempts were not focused on whole organs but rather on cells and tissue. The first recorded xenotransplantation attempt occurred in 1667, when physicians Jean-Baptist Denis and Paul Emmerez transfused the blood of a lamb to a 15-year-old boy in Paris. In a key milestone of

xenotransplantation Mathieu Jaboulay and Alexis Carrel introduced vascular anastomosis, or the process of connecting blood vessels to each other in 1912. This provided a means to restore blood flow in transplanted organs. After this, many scientists and surgeons attempted xenotransplantation primarily from nonhuman primates but also from other mammals like pigs, sheep, and even rabbits [5]. The results of many clinical trials were disappointing because of a lack of immunosuppression, and most grafts failed due to HAR. There was thus a 40-year hiatus on xenotransplantation experiments because scientists deemed the procedure too dangerous without sufficient immunosuppression. [5].

When immunosuppressants were pioneered in the 1960s, medical attention to both allo- and xenotransplantation was increased and clinical trials saw the longest survival times yet. Likely the most famous instance of xenotransplantation occurred in 1984, when Leonard Bailey attempted to implant a baboon heart into a 12-day old infant now known as "Baby Fae," who was born with a life-threatening heart condition. The operation was successful, but the baby died 20 days later due to antibody-mediated rejection (AMR)[5]. It was during the 1990s that scientists realized they had to make alterations to the transplantation process beyond simply adding immunosuppressants. By genetically modifying pigs, scientists began exploring new ideas for improving xenotransplantation. In 1992, the first genetically modified pig, named Astrid, was modified with the human decay-accelerating factor, a protein that inhibits complement activation in humans, the system that enhances the ability of antibodies to attack foreign particles. Over the last couple decades, the invention of new technologies such as cloning, CRISPR, and TALENS have improved the ability to genetically alter pigs [5].

The largest advancement in genetic modification in pigs was the removal of the Gal antigen from pigs. Because this is the biggest component in graft failure, *α*1,3-galactosyltransferase gene-knockout (GTKO) pigs were engineered. Even though Gal was no longer present in the genetically modified pigs, evidence of a non-Gal antibody response emerged, resulting in further genetic remodeling that will be discussed in the next section. Complete removal of the Gal antigen was accomplished in 2003, another significant step that allowed mitigation of HAR and survival of pig-to-baboon hearts for 6 months and pig-to-baboon kidney transplants for 2-3 months [5]. The persistence of perioperative cardiac xenograft dysfunction (PCXD), however, became an important barrier as it is unrelated to immune response and thus only emerged in the presence of immunosuppressants. As clinical trials progressed, new innovations in preservation techniques were shown to mitigate PCXD [5].

In recent years, researchers have learned a multitude of lessons regarding AMR from the field of allotransplantation. HAR has always been the main obstacle for both allo- and xenotransplantation-- scientists have almost eliminated the risk of hyperacute AMR in allotransplant; recent advances in xenotransplantation have been based on clinical trials in allotransplantation. For example, testing for anti-alpha-1,3-galactose IgG and IgM and IgE antibodies can be useful for preventing AMR. Monitoring methods from allotransplantation can also carry into xenotransplantation-- closely watching the activity anti-pig donor-specific antibodies are critical and thus need to be standardized during transplants [6]. Scientists have also learned from ABO blood type incompatible transplants that it is possible to desensitize patients' immune systems when there are antibodies present. Clinical trials examining these types of transplants reveal that outcomes are comparable between ABO incompatible and compatible transplants, which can be applied to renal xenotransplantation trials [6].

The exploitation of genome-editing technologies in xenotransplantation

There has been significant progress in the field of xenotransplantation with the emergence of CRISPR/Cas9 genome editing. The anti-Gal antibody can be suppressed using typical immunosuppressants, but as

mentioned earlier, this puts the immune system at a much greater risk of infection. Hence, scientists have turned their efforts towards the strategy of genetically engineering donor pigs for resistance to rejection [7]. Instead of suppressing the immune system's ability to attack the Gal antigen, researchers demonstrated in a 2005 heart transplant study that by genetically engineering pigs without the Gal antigen known as GTKO, HAR was eliminated because the Gal antibody had no effect [7].

CRISPR/Cas9 is the most promising method of genome editing because it is simpler than other mechanisms and for its efficiency in gene targeting. CRISPR specifically removes foreign genetic elements and only requires a very short guide RNA targeting sequence of less than 20 nucleotides, in contrast to other methods such as TALENS, which requires an array of tandem repeat molecules. CRISPR is also extremely efficient in targeting multiple genes at a time and can therefore make several genetic modifications in a single animal [4].

So what genetic modifications are actually needed to overcome the barriers to xenotransplantation? Recall that the main problem is cross-species incompatibility and HAR, which is caused by the presence of Gal epitopes on the surfaces of pig cells. It has been found that the preformed anti-Gal antibodies account for 1% of the total antibodies that circulate the human immune system [4]. The enzyme GGTA1 is responsible for the synthesis of Gal-- it is found in most mammals except humans and NHPs. In 2002, two groups achieved complete removal of the enzyme GGTA1 through genetic engineering, producing the first GTKO pigs. HAR, however, is also caused by human complement systems. Because porcine complement-regulatory proteins (CRPs) cannot control the human complement system, the complement cascade causes damage to pig grafts. Scientists solved this problem by modifying pig organs to express human CRPs rather than porcine ones [4].

However, more genetic modifications are not always better for a successful transplant. Extensive genetic engineering can lead to decreased animal viability and malformations. Currently, the latest available pig only has 10 genetic modifications [6].

Advances in cardiac xenotransplantation

On January 7, 2022, David Bennett, a 57-year-old patient with terminal heart disease, was given a genetically-modified pig heart in a groundbreaking first. It is a landmark in cardiac xenotransplantation because the study demonstrated that a genetically modified pig heart can function without rejection from the body. E. Albert Reece, the Vice President of Medical Affairs at UM Baltimore, said that the "unprecedented and historic procedure highlights the importance of translational research which lays the groundwork for patients to benefit in the future. It is the culmination of our longstanding commitment to discovery and innovation in our xenotransplantation program [8]." While Dr. Reece is correct that the surgery was unprecedented, it was also informed by decades of research on cardiac xenotransplantation. This procedure was an orthotopic one (meaning that the transplant took place in the organ's normal place in the body), the likes of which have previously been impossible due to PCXD and ischemia, but

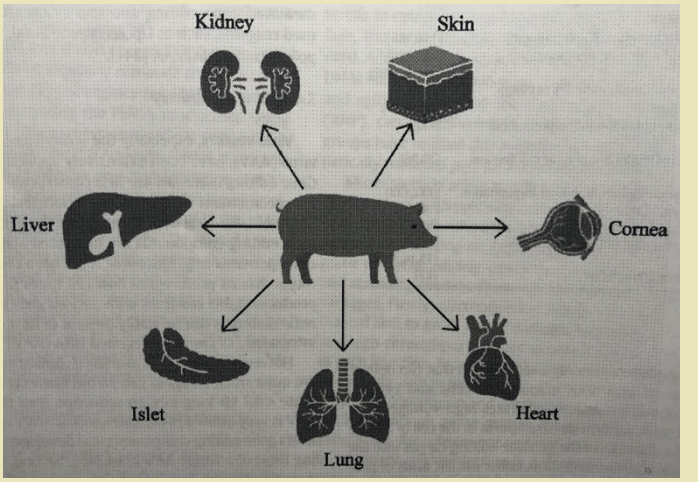
Surgeons at the University of Maryland prepare the porcine heart for transplantation [8].

despite this the operation was successful due to improved methods of organ preservation and post-transplant monitoring technology. Furthermore, the transgenic expression of membrane protein CD39 in donor pigs improved graft survival in the latest procedure [7]. Scientists deactivated 4 porcine genes, including Gal, and inserted six human genes to make the heart more tolerable to the immune system. Unfortunately, the operation was not actually a study-- it was an emergency treatment that was given last minute approval by the FDA, making information regarding the details difficult to find. However, Bennett died two months after the surgery due to unknown causes. Scientists suspect that it was not related to graft failure and plan to publish a clinical study in the near future detailing the results of an investigation into the patient's death. Nevertheless, it was an incredibly exciting demonstration of xenotransplantation that proves the toil of the last 50 years has been worth it. Through the media, it also made the public more aware of the field of xenotransplantation, a definite goal of the operation.

Advances in renal xenotransplantation

Surgeons at the University of Alabama at Birmingham reported on January 20, 2022, that they had for the first time successfully transplanted kidneys from a pig to a brain-dead human model. It was also the first time a pig-to-human xenotransplantation study has been published in a peer-reviewed medical journal. This latest procedure followed all the steps of a human-to-human transplant, according to the New York Times. The surgery was carried out in hopes that it would lay the groundwork for a small clinical trial with live patients that the head surgeon, Dr. Jayme Locke, hopes to begin by the end of 2022, according to the New York Times [3]. Because 50% of kidney failure patients in low-income countries currently die due to the lack of access to dialysis, kidney xenotransplantation clinical trials can be justified if they could guarantee patient survival at better rates than dialysis. Kidney transplantation has been a large focus of the xenotransplantation community because of its less severe consequences in the case of kidney vs. heart graft failure. In the past two years, various pig-to-NHP transplants have emphasized the importance of $CD4^+$ T-cell depletion, a process during which helper lymphocytes are destroyed in order to weaken the immune system [6]. This process is in fact how HIV weakens the immune system-- but in kidney transplantation, it is a necessary step to prevent xenograft rejection. Scientists have also experimented with different GalT-KO pigs in recent years to add more human transgenes. This has proved to increase biocompatibility and higher safety in the presence of PERVs [6]. In the Birmingham study, pigs known as 10-GE pigs were provided by biotechnology company Revivicor, a sub-branch of the company that carried out the Dolly the Sheep cloning experiment in 1996. The pigs have 10 genetic modifications that comprise of two human complement inhibitor genes (hDAF, hCD46), two human anti-coagulation transgenes (hTMB, hEPCR), and the two immunomodulatory genes of hCD46 and hHO1, which suppress their respective roles in the immune rejection process. These modifications also included the knockout of 3 pig antigens (including Gal) as well as the pig growth hormone gene to prevent the organ's cells from increasing size post-transplant. Interestingly, the 10-GE pigs do not express red blood cell antigens ABO and therefore are universal blood donors. In this latest study, scientists did not observe any signs of hyperacute rejection, proving that the genetic modifications in 10-GE pigs were sufficient to prevent HAR [2]. The kidney functioned normally, producing creatinine and urine and no transmission of PERVs was evidenced. Another goal of the study was to evaluate the human decedent model compared to the NHP model. Surgeons found that while the brain-dead model has numerous limitations, it is capable of pointing out additional areas where more understanding was needed. Therefore, the model has potential, according to the authors of the study, to "propel not only the field of xenotransplantation forward but to answer a multitude of other scientific questions unique to the human condition [2].

Conclusion: on the ethics of xenotransplantation



The pig organs being used in preclinical trials [4].

Many scholars in the fields of science and philosophy have raised questions about the ethical issues occasioned by xenotransplantation. Most ethical problems raised by transplantation in the past have been related to the allocation of scarce organs, but if the enormous potential of xenotransplantation is realized, it brings with it another set of ethical dilemmas. While pigs have been selected as the ideal species for human organ transplantation because of reasons previously listed, their use for clinical trials has been objected to by various animal rights activists due to the animal's intelligence, sentience, and capacity for suffering [9]. There are also problems with the procedure itself: the patient must offer consent to lifetime surveillance and also to the risk of contracting PERVs, which scientists have still not found a way to genetically-engineer out of pigs. Therefore, scholars have suggested that the surgeons and doctors administering the procedure should present the organ recipient with necessary information regarding the operation and post-procedure risks [9].

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4. Musical Intervention: Innovative Autism Therapy

Sarah Thomas, ‘22

Music can be a powerful tool for people with Autism Spectrum Disorder (ASD). ASD is characterized by difficulties in social interaction, communication, and repetitive behaviors [1]. Researchers have observed exceptional artistic abilities in those who have autism, especially in the ability to process music with energy, emotion, and focus [2].

Music as a Strength

Many people with ASD identify music as a strength during their childhoods. Singing, for example, can promote early speech in children by accessing neurological pathways that correlate to improved social communication. Another characteristic of ASD is pattern recognition abilities, making rhythmic sequencing and memorization easier than it may be for a neurotypical person [1].

Effects of Music on Social Communication

Whether it is listening to a song with complex and intricate rhythmic patterns, or tapping a simple beat on a tambourine, any form of music processing stimulates the brain. Many children with ASD even respond better to music than they do to speech in fronto-temporal brain regions. They also often exhibit enhanced music skills such as perfect pitch, melodic memory, and emotional responsiveness to music. These abilities carry over to the positive impact of music on social skills in young children. Musical interactions can improve emotional empathy and bonding, and neuroimaging studies show that participating in such activities engages a multimodal network of brain regions responsible for hearing, movement, emotion, pleasure, and memory [4].

Music Intervention Therapy and Music Education

Many families engage their child with ASD in music therapy due to its positive outcomes on social communication and auditory-motor connectivity. This therapy often includes structured improvisation, vocalization and singing, and listening to music, both pre-recorded and live. These nonverbal musical interactions develop learning skills, communication, and emotional expression. Music education is also a common building-block for children with the disorder to experiment and find an interest in general music.

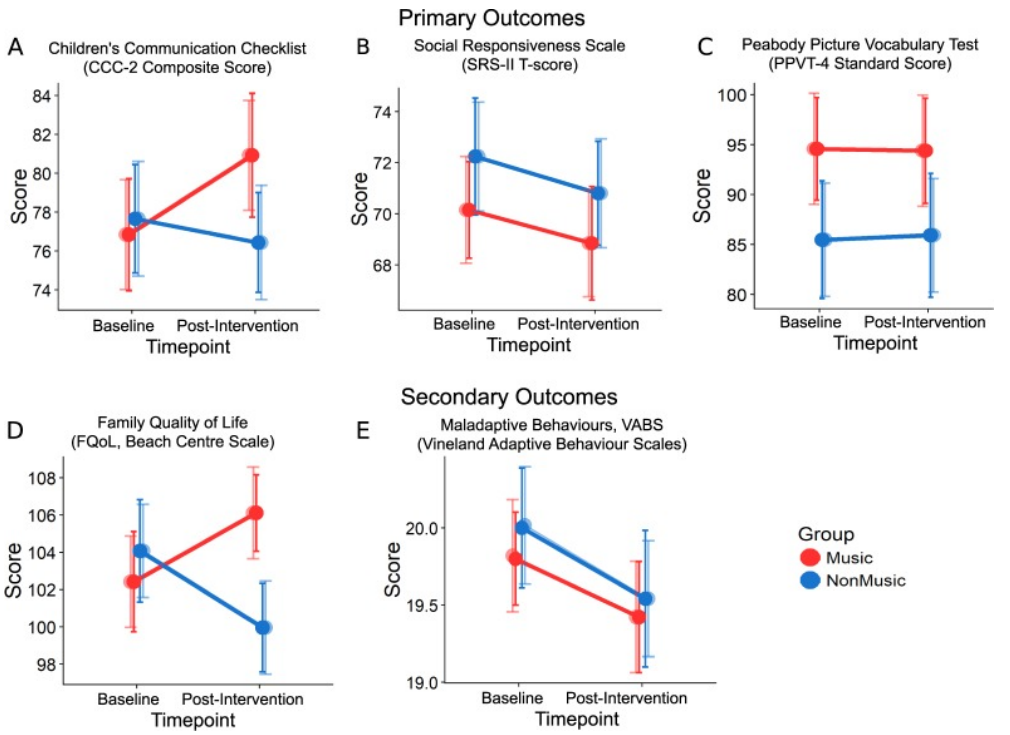
While music education in neurotypical populations is often attributed to learning how to play an instrument, read music, and study music history and composition, it is utilized especially in neurodivergent populations because it has been proven to be more effective than visual art education. Visual art education fails to stimulate the mechanisms that overlap verbal- and music-processing functions in the brain, which leads to greater improvements in language skills. Music education for neurodivergent children and adolescents also results in brain plasticity due to the enlargement of brain areas including the auditory cortex and limbic areas [3]. An excess of neural conductivity is found to be a contributing factor in ASD, so a reduction in neural conductivity may be beneficial in this condition.

Montreal Neurological Institute Study

The trial consisted of fifty-one children with ASD, aged 6-12 years. This set of children was randomized, providing some with 8-12 weeks of music intervention and others with 8-12 weeks of non-music control intervention. The study took place from April to December of 2016 through the Montreal Neurological Institute at McGill University. Prior to the study, assessment scores were recorded for the participants, including the Children's Communication Checklist and the Vineland Adaptive Behavior Scales (a subscale for maladaptive behaviors). These same assessments are taken after the 8-12 weeks to evaluate the effects of music intervention on aspects of the disorder such as communication and maladaptive behaviors [4].

Results of Music Intervention Therapy

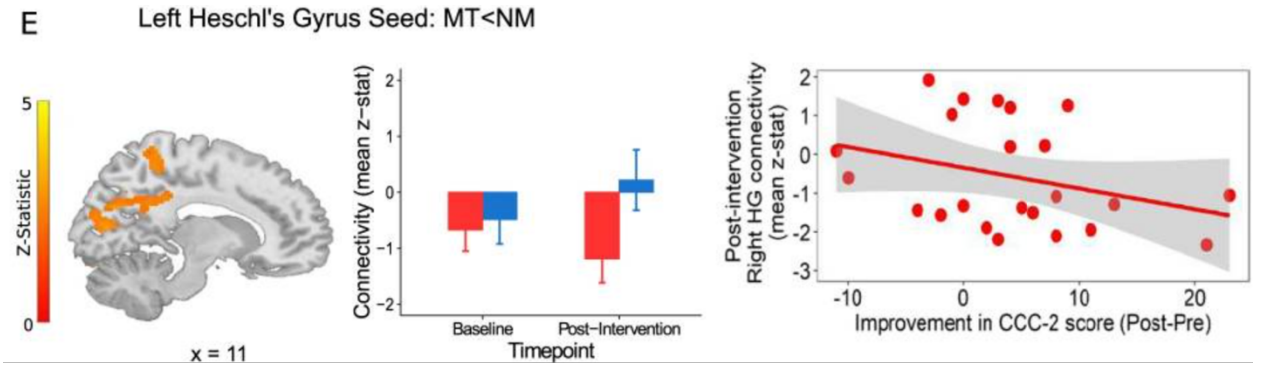
Using a mixed model, the study proved that music therapy results in enhanced communication, as the scores received on the Children's Communication Checklist improved, with a mean difference of 4.84. Neuroimaging data showed a decrease in brain connectivity correlating with improved communication post-music intervention. The Vineland Adaptive Behavior Scales—measuring communication, daily living skills, socialization, motor skills, and maladaptive behavior—also detected a decrease in maladaptive behaviors after the 8-12 weeks [4].



[6]

Effects on Brain Connectivity

8-12 weeks of music intervention therapy can alter brain intrinsic connectivity and improve social communication in school-age children with ASD. Due to the improved scores on the Children's Communication Checklist and Vineland Adaptive Behavior Scales, as well as decreased brain connectivity in over-connected regions of the brain, music therapy can be proposed as a strength-based rehabilitation tool for children on the autism spectrum [4].



[7]

Efficacy of Music Intervention Therapy

Music intervention therapy's proven success in helping children with ASD with communication skills and improved brain connectivity confirms its efficacy as a treatment of the disorder. Families of children on the spectrum have continued to consider music therapy when seeking treatment for the disorder, due to its positive effects. While being on the autism spectrum entails many limitations, utilizing the strength of music-processing to benefit the behaviors and interactions of people with autism is necessary for fostering social communication skills, while simultaneously promoting their natural abilities.

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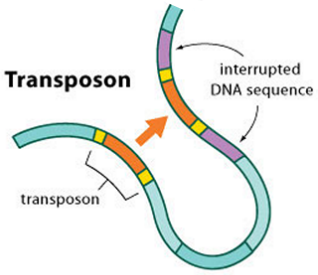
5. The Corpse Flower: The Stinky Flower that has Changed Our Outlook on the Future

Tanvi Gorre, ‘23

The genome governs life on earth. It is the blueprint to make life and is constantly evolving. Through changes in its instructions, the genome is able to create organisms with new abilities or even birth different species. A plant that has just changed our understanding of evolution is the corpse flower, a parasitic plant species that is widely described as smelling like rotting flesh.

This potent flower only blooms every seven to nine years, making it a spectacle to watch come into its full form (the power of exclusivity). Beyond its horrid smell and short bloom, it is the subject of a major scientific breakthrough. Scientists at Harvard sequenced the DNA, the genetic material of the corpse flower, and made several shocking discoveries that have major botanical implications for the future: they found that the corpse flower acquired many genes from their host through transposons [1].

A transposon is a piece of genetic code that is able to move throughout the DNA, which is why scientists describe them as jumping genes. These facets of genetic code serve several functions depending on their class of transposon. A Class II transposon codes for transposase, which is an enzyme that cleaves the transposon from the genome so that it can move elsewhere in the genome [2]. Retrotransposons are very similar to Class II transposons, except they leave the original copy of the transposon, and their copies then occupy other parts of the genome [2]. These small pieces of genetic information are important because the movement of transposons can cause mutations and chromosome rearrangements—the backbone to evolution [1].



This is a representation of a transposon in DNA

Mutations caused by transposons can change an organism on a fundamental level. DNA is used to make proteins, a macromolecule that makes up some of the defining characteristics of species. When mutations cause a member of a species to have different characteristics, that organism becomes a new species [5]. Scientists state that if two plants are unable to produce a fertile offspring, then they are not the same species [6]. Consequently, this means the corpse flower could have potentially produced hundreds of species and will contribute more in the future.

Having disproved many previous ideas about how transposons work, the corpse flower has sparked questions in the scientific community about the volatility of transposons and how big a role they play in evolution. Transposons could potentially have moved genetic material between hosts and parasites for millennia. This discovery has introduced the possibility that many of the parasites on earth (which account for nearly 40% of all life) may actually be able to change the course of evolution [4]. Somehow, a flower that smells like the dead has given us hope for new life in the future.

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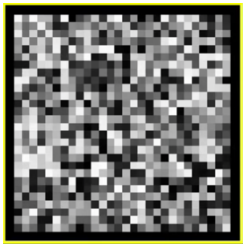
6. Digital DaVinci: Artificial Intelligence and Art

Tom Zhang, ‘23

Machine learning (ML) and artificial intelligence (AI) are usually seen as only capable of making basic, calculated decisions; examples like classifying facial emotions as sad or happy, or detecting objects appearing in images generally come to mind. This is true in many cases: the traditional neural network (NN), a type of ML model based on summing multiples of linear functions (i.e. w=ax+by+cz), can recognize handwritten digits, but could never draw a digit. Take the example of Grant "3Blue1Brown" Sanderson's neural network for digit recognition: it has about 98% accuracy in terms of recognizing handwritten digits it has never seen before, but if you were to ask it the number that an input of random noise looks like, it may confidently output a five [1].



Noisy Input [9]

Not great, right?

Competition Breeds Success

Since optimization in NNs focuses on minimizing the difference between the correct answer and the answer chosen, training focused on picking simple multiple-choice questions does not result in a creative or nuanced model. If a low-confidence model took an approach of choosing conservative probabilities for each of the ten digits, the error would be very high for the correct digit and the model would be forced to become more confident. So, the goal, in effect, is picking one correct answer confidently and outputting low probabilities for all the other digits.

However, with a change in the structure of the model, an emergent sense of creativity can arise from the optimization of decisions. In the real world, when students learn to become better writers, they need mentors to give feedback on the quality of their essays and stories. Similarly, in the architecture of general adversarial networks (GAN), one network, the generator, is responsible for making new fake content, while the other network, the discriminator, is responsible for judging whether the creation is real or fake [2]. Both networks improve simultaneously; once the generator can trick the discriminator into thinking that its content is real, the generator can be used to make new art, stories, or other media.

GANs apply to many problems, especially applying unique filters to images, known as image-to-image translation. Variations in structure allow GANs to solve more problems, including winter to summer, horse to zebra, and night to day.

However, as some may have noticed, this does not prove that GANs are capable of creating something novel. Covering all the mountains and trees with snow and making the sky a little darker makes a good tool for editing photos, but not a good artist. In order to emulate the unique styles of skilled human artists while maintaining originality, the GAN model must be restructured.

Input and output images by a winter-to-summer GAN [10]

Reconstruction and Progression

Originally, the discriminator only cared about whether the image it saw could be considered art. In contrast, to accomplish the goal of having a unique style, the creative adversarial networks (CAN) introduce a second aspect to consider for the discriminator: stylistic categories [3]. For example, if the goal were to generate paintings that do not belong to any period, then if the discriminator finds that a painting's style is too similar to that of an existing artist, the generator would be punished for it. Note that the CAN can still be applied to emulate the style of a single artist. For example, to emulate the general style of Picasso, one could punish the generator for creating a piece that is too similar to Picasso's Blue Period. In essence, the defined categories dictate what the model can create.

Although a looming weakness of art-generating models is that there is no absolute way to quantify whether a model can be improved, there is always an infinite number of optimization functions to be tried, but we will never know which is best.

Furthermore, what someone views as an imitation, someone else may view as a masterpiece. Beginning in 2018, Christie's, a world-renowned auction company, has held an annual Art+Tech Summit [4]. At the first summit, on October 24th, *Edmond de Belamy*, a painting by a GAN, sold for $432,500 [5]. For those aware of the popularity of the non-fungible token (NFT), it should be no surprise that art has become increasingly intertwined with technology.

Edmond de Belamy, from La Famille de Belamy, published by Obvious Art

Improvements

In April 2022, massive strides were made in the field of art-generating AI with a greater emphasis on interaction with the user, allowing anyone to create anything that they can imagine. With OpenAI's new model, DALL·E 2, users can input a text description of an image and the model will output many interpretations, with variations in their composition and details. For instance, given the description, "An astronaut playing basketball with cats in space as a children's book illustration," DALL·E 2 will generate the following images, among others:

An astronaut playing basketball with cats in space as a children's book illustration, created by DALL·E 2 [12]

Impressive, right? However, DALL·E 2 is not a GAN, although it is similar in purpose. Instead, the model was trained on images and their descriptions, learning the characteristics of each object and also how objects relate to other objects or actions [6].

As for the specific architecture, the model first converts the caption into a text embedding, which is like a set of coordinates that describe the text numerically. Then, the text embedding is mapped to an image embedding, which incorporates some variance so that not every final image is the same. Finally, the image embedding is decoded into an image, which then gets upsampled until its resolution is 1024 pixels by 1024 pixels [7].

According to to OpenAI, DALL·E 2 "helps people to express themselves visually in ways they may not have been able to before" and "helps humans understand how advanced AI systems see and understand our world" [6]. Both of these missions are crucial to people living in a world in which AI is rapidly integrating into various industries.

Conclusion

Ultimately, AI is a constantly evolving field with major changes taking place every day, with every single change filtering down to other industries. Yet, like applying paint to a canvas, there are always new ways to apply AI to real-world problems, introducing human creativity into the sterility of robotics. In the future, there will definitely be models which build off of GANs and DALL·E 2 which may be able to revolutionize architecture, engineering, and science as a whole. As Leonardo Da Vinci once said, "Study the art of science. Study the science of art. Realize that everything connects to everything else."

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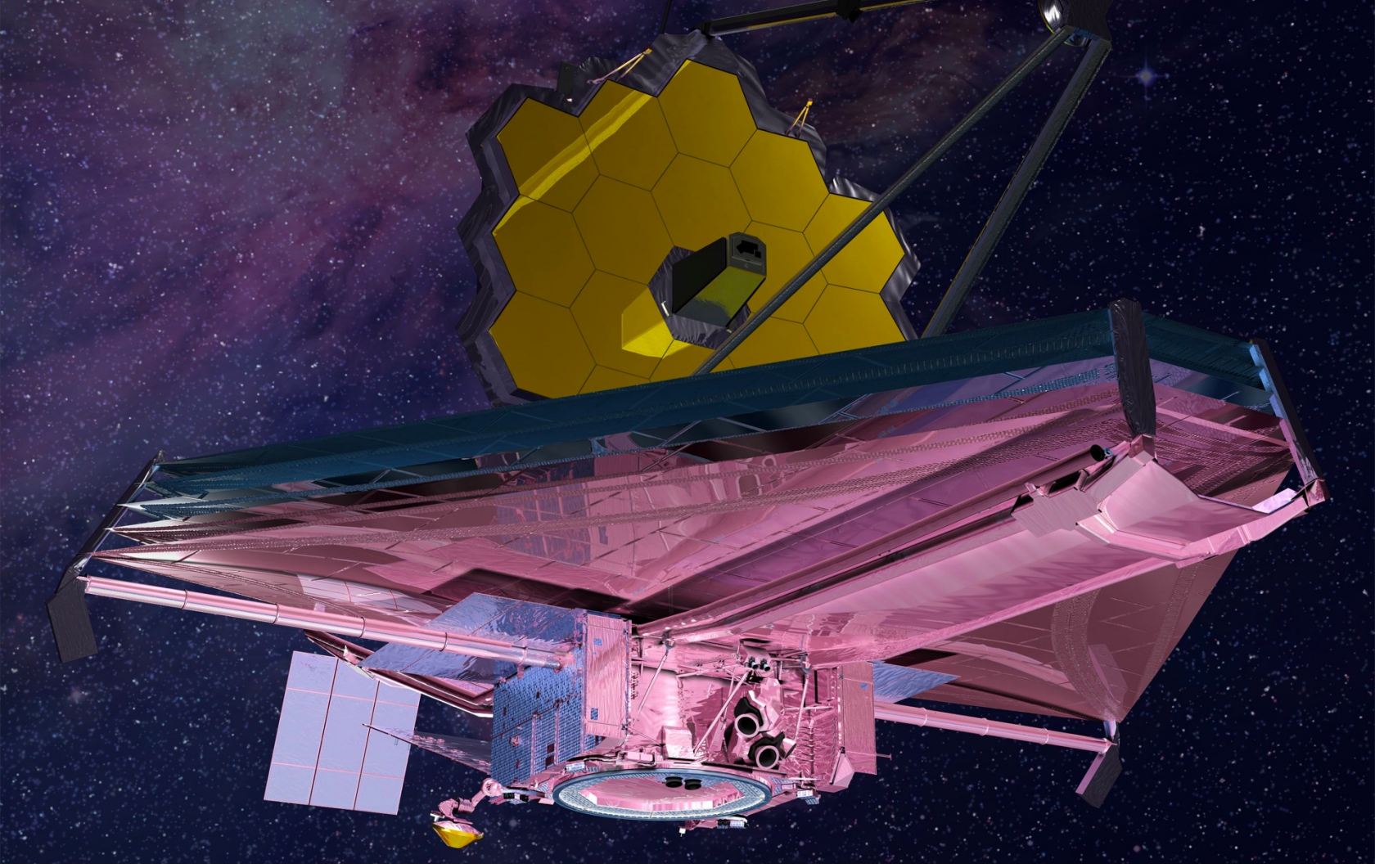
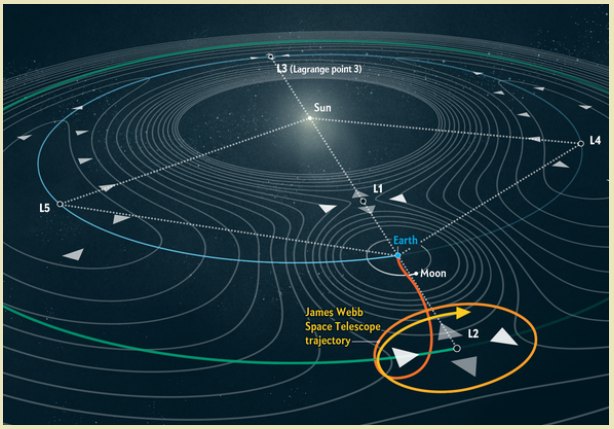


7. From French Guiana to the Beginning of Time: The James Webb Space Telescope's Journey to Help Us Understand Our Universe

Will Boberski, '25

On Saturday, December 25th, 2021, at 7:20 a.m. ET, the James Webb Space Telescope (JWST) lifted off from the European Space Agency (ESA) Spaceport in Kourou, French Guiana and began a journey to the beginning of time [2]. At a point 1.5 million kilometers away from Earth, the JWST will view and photograph exoplanets, galaxies, and some of the universe's first stars, using cutting-edge technology to push the boundaries of astronomy and further our knowledge of the cosmos [3].

Twenty-seven minutes after liftoff, the JWST separated from an Ariane 5 rocket and began its solo flight to the second Lagrange point (L2) [3]. Lagrange points are positions in space where objects tend to stay put relative to two large objects on either side, balanced between the gravitational tugs from the two opposing masses [4]. The JWST will orbit L2 to reduce fuel consumption and observe deep space, while remaining in front of the Sun, Earth and Moon [5].



Lagrange points, including L2, which the JWST will orbit, are regions in space where the gravitational pull on an object is balanced [6].

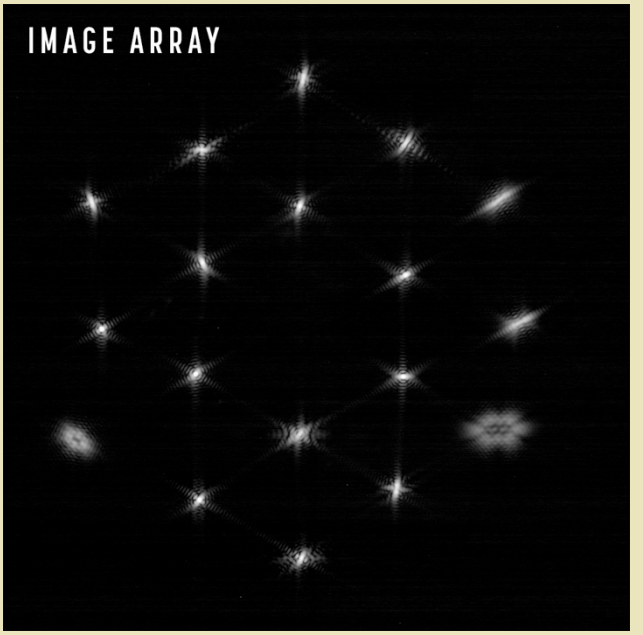
The JWST's primary mirror is packed in preparation for launch [7].

After the Hubble Space Telescope was launched in 1990, a 1996 committee recommended that NASA should next develop a telescope to orbit beyond Earth's moon, with a mirror larger than four meters in diameter to detect infrared light [8]. The construction of this "Next Generation Space Telescope," as it was known at the time, began in 2002 and cost more than $10 billion USD to complete due to repeated delays [9]. After decades of hard work, the JWST finally launched in 2021, and viewed its first star on February 11, 2022 [10].

The JWST's Technology

The JWST's main mirror, 6.5 m in diameter, is made up of 18 hexagonal gold-plated beryllium segments [4]. At the start of the JWST's month-long alignment process, each of the mirror segments acted like its own telescope. The JWST's first image of space thus contained 18 images of the same star from each of the 18 mirror segments [10]. The next stage of the alignment process required engineers to align the mirrors corresponding to each of the 18 different dots of starlight into a hexagonal formation using delicate motors mounted behind each mirror segment. The JWST's mirror segments will eventually align fully so that the 18 separate stars will "merge" into one focused and detailed image [11].

The JWST's first image: a single star viewed independently by each of the telescope's 18 mirror segments [10]



The JWST team uses precise sensors and motors to guide the mirror's 18 segments into their proper hexagonal shape [11].

The defining feature of the JWST is its ability to detect near-infrared and mid-infrared light. Hubble was designed to detect ultraviolet light, visible light and parts of the infrared spectrum. In order to detect infrared light from distant stars, the JWST's telescope must be exposed to as little infrared light from the Sun as possible. The JWST telescope can only operate properly at a temperature of -233 degrees Celsius; otherwise, its sensors are overwhelmed. To block light from the sun, the JWST has a tennis-court sized sunshield, made of five ultra-thin layers of Kapton E plastic coated with aluminum and silicon [12].

In addition to the main telescope, the JWST also carries a Near-Infrared Spectrograph (NIRSpec), a Near-Infrared Camera (NIRCam), a Near-InfraRed Imager and Slitless Spectrograph (NIRISS), and a Mid-Infrared Instrument (MIRI) [3]. These instruments will use spectroscopy to find the temperature, mass, and chemical composition of stars, dust clouds and exoplanets. Spectrographs break up light into different wavelengths, so as to identify specific chemicals that emit light at certain wavelengths. However, the MIRI only operates properly at a temperature of -266 degrees C, necessitating a cryocooler refrigeration system[3].

This additional refrigeration system presented a unique set of engineering challenges. The cryocooler needed to contain as few moving parts as possible, in order to run reliably for long periods of time and minimize the risk of moving the telescope out of focus. The cryocooler also needed to use less than 2,000 watts of electricity, the amount produced by the JWST's solar array [4]. To solve these problems, NASA and the ESA, who collaborated on the MIRI, created a closed-loop system in which refrigerant is continually reused. A transducer inside a closed resonator chamber filled with gas uses a thermoacoustic standing wave to create a pressure and temperature gradient. The hot and cold areas of the tube are separated by a porous stack. Heat exchangers are placed on either end of the stack, routing the "hot" molecules out into space and the "cold" molecules to the MIRI [4].

The JWST's Goals

What does the JWST aim to see, and why is it important? The JWST will help astronomers understand what the early universe looked like, how the first stars, galaxies and black holes formed and evolved, the lifecycles of stars, how planetary systems form and evolve, and whether the atmospheres of various exoplanets contain the building blocks of life [3]. The JWST will also be able to detect light from the early universe.

For 370,000 years following the Big Bang, the universe was filled with a dense fog of ionized gas [13]. As the cosmos expanded and cooled, subatomic particles combined to form the first neutral atoms [13]. The formation of atoms allowed heat from the Big Bang to travel freely. We see remnants of this heat today as cosmic microwave background radiation, its wavelengths stretched, or redshifted into the infrared range, by the universe's expansion [13]. Due to the absorbing properties of atomic hydrogen, the universe became opaque at shorter wavelengths [13]. This initiated a period known as the Dark Ages, so named because of the absence of stars and the dense hydrogen gas that filled the cosmos [13]. Over time, regions with higher densities of gas began to condense under gravity [13]. Neutral matter clumped, cooled and collapsed, leading to nuclear fusion and the formation of the first stars and galaxies [13]. These short-lived but massive stars converted hydrogen and helium into oxygen, iron and other elements [13]. Additionally, their radiation heated the surrounding medium and reionized hydrogen [13]. Once most of the universe had been reionized, around one billion years after the Big Bang, wavelengths of light across the electromagnetic spectrum could travel freely [13]. Redshifted by the expansion of the universe, this first light now has long, infrared wavelengths, and will, hopefully, be detected by the JWST.

The JWST's launch and the beginning of its telescope alignment marks the start of a new era in astronomy. The telescope's massive mirror and specialized instruments will help scientists across the world to better understand the cosmos and peer back through time itself. Although the JWST might not have, thus far, attracted the publicity of the Hubble Space Telescope's images or the Mars *Perseverance* rover, the JWST's contributions to science, and our understanding of our vast and beautiful universe, will be immeasurable.

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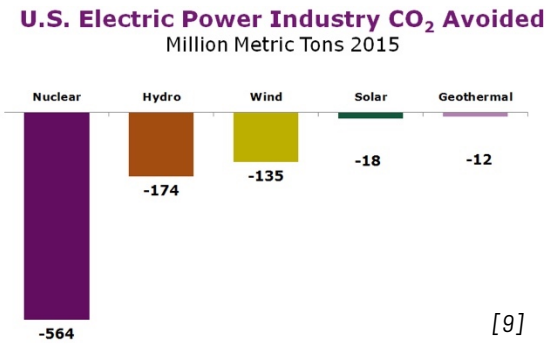
8. Molten Salt Reactors: Nuclear Energy's Best Chance at Redemption

Will Heisler, '22

The Fallbacks of Current Nuclear Energy Production:

Seemingly dismissed in current conversations regarding the future of global energy production, nuclear energy has fallen heavily out of favor in recent decades. Yet, solid core reactors have represented an ever-present backbone of US energy production since their largest implementation in the 1970's, and they continue to supply 18.9% of America's total energy output through a mere 56 plants nationwide [2]. This figure accounts for 52% of energy produced from carbon free sources [2]. The potential of nuclear fission as a clean energy source is one that cannot be overlooked in a time when an overhaul of global energy grids has reached its utmost importance. Yet, glaring past failures and current drawbacks have entirely spoiled public perceptions of nuclear energy. In order for trust to be restored in this highly dependable and efficient source of renewable energy, the technology needs to experience industry-redefining innovations. A solution may have been found in an advanced new generation of nuclear reactor known as a molten salt reactor (MSR). MSR's could very well be the key to implementing cheaper, safer, and more effective nuclear power.



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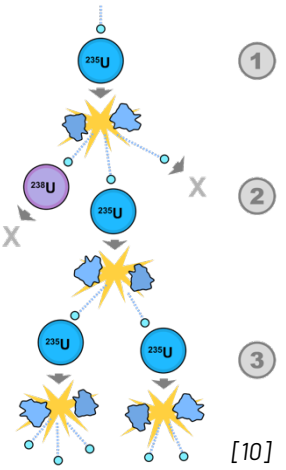
Molten Salt Reactors:

Traditional reactor technology relies on a solid uranium fuel in order to produce energy through nuclear fission. Within the core, these electrostatically unstable atoms of Uranium-235 eject neutrons from their nuclei in the form of radiation [5]. These particles slam into other uranium atoms and the resulting collision splits their nuclei apart. This causes an immense expulsion of energy and the release of more neutrons that impact other nearby uranium atoms. This chain reaction of atomic fragmentation is carefully moderated within the reactor core and the thermonuclear fission process releases this energy in the form of heat. A

standard nuclear power plant operates at temperatures nearing 300 degrees celsius [4]. Water, pumped around the reactor core, is vaporized by this heat and the resulting pressurized steam is utilized to turn electricity-producing turbines. Molten salt reactors differ from this conventional technology; rather than solid uranium fuel rods, MSRs utilize molten liquid fluoride or chloride salts to facilitate this chain reaction [4]. These salts are combined with an alternate radiation source like Thorium to produce a fully liquid reactor core in which the heat of the fission reaction continuously maintains the molten state of the salt [5]. In order to begin electricity production, the fuel itself is pumped out of the main reaction chamber. It then travels through the piping of a heat exchange chamber where surrounding water is vaporized. In a process identical to conventional reactors, the high pressure vapor drives turbines that produce the final electrical load.



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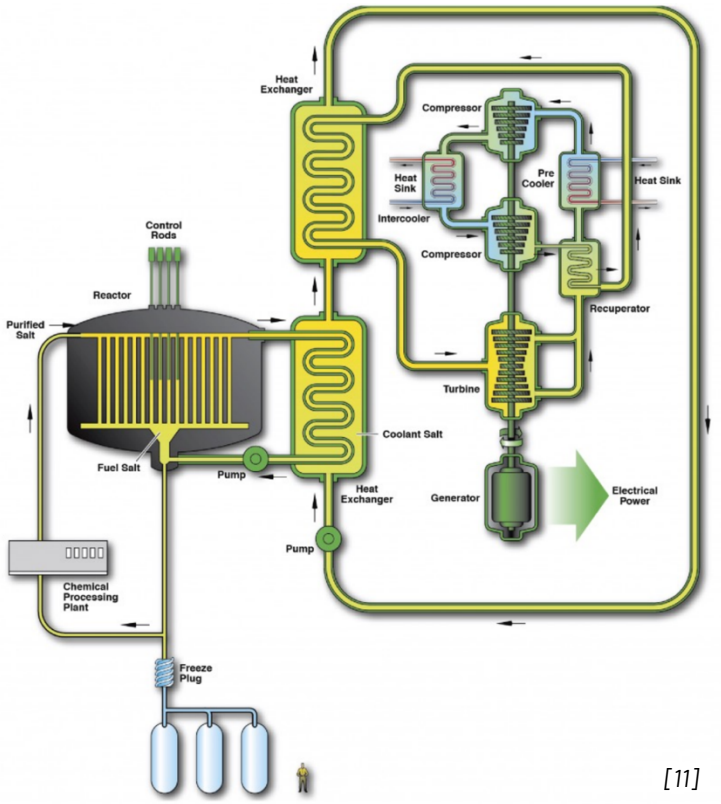

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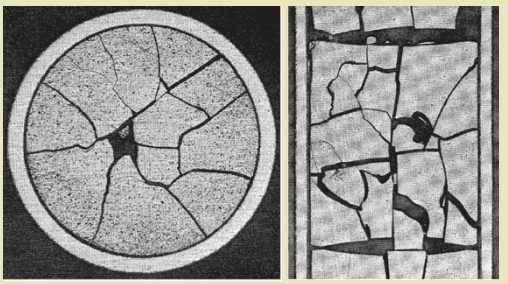
Advantages Over Current Nuclear Technology:

While still an extremely effective source of clean energy, current reactors do feature drawbacks which have dissuaded more widespread development. The nature of a solid radioactive core means that as atoms of fissile U-235 are split, resulting byproducts remain within the fuel and reduce its potency as the reaction occurs [1]. As a result, this material must be removed and discarded in the form of nuclear waste. Additionally, concerns over reactor safety following the accidents of Chernobl, Fukushima, and Three Mile Island continue to harm public confidence in this energy source [7]. MSR technology brings solutions to each of these problems along with other key advantages that result from their liquid core design and the potential use of thorium isotopes.

When discussing nuclear energy, the criticism of waste is often the first raised against this technology. The creation of non-fissile by-products such as Xenon have a poisoning effect on the reactor core and pose the risk of cracking the fuel rods if enough gas becomes accumulated within the reactor chamber [1]. These cores therefore must be discarded despite a high percentage of the necessary uranium remaining un-reacted. Being unable to separate the U-235 from these other radioactive actinides, the fuel rods must be disposed of entirely [8]. The utilization of a liquid state core offers the ability to remove these parasitic atoms of Xenon during reactor operations as the gas can simply bubble up through the molten salt to be vented out of the fuel circulation [8]. By-product removal of this fashion allows nearly all of the fissile actinides to be consumed by the reactor. Therefore, it can operate on an extended timeline and leave a waste product that will remain radioactive for orders of magnitude less time than standard waste.



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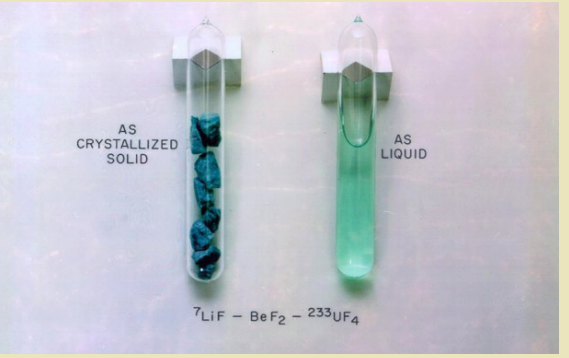
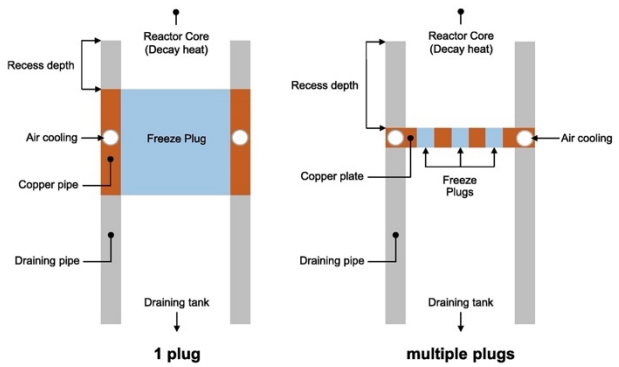
The use of molten fluoride salts as a fission medium creates the ideal environment to utilize alternative sources of neutron emitting isotopes. Proposed Uranium replacements include Thorium, an element that produces far more stable post-fission by products. Perhaps even more promising is the potential to utilize stored stockpiles of nuclear waste [3]. Since solid core reactors are only able to convert a mere 4% of each pellet's total Uranium content, an MSR's more complete fission process allows this waste to be recycled back into the reactor for further use [7].



Fluoride fuel for a molten salt reactor [7]

There are also immense safety benefits to be gained by utilizing a molten reactor core as opposed to a solid fuel assembly. Most presently, the catastrophic event of a "meltdown" which has occurred during previous nuclear disasters, would not be possible in this format considering the already liquid state of the fuel [7]. If the need did arise for a reactor to be shut down, this process can be completed by melting a freeze plug and draining the molten salt into a holding receptacle, immediately halting the fission reaction and further steam buildup. This shutdown process can be completed without the need of boron control rods which serve as neutron absorbers and the last-resort fail-safe in current reactor models [2]. These control rods are effective in current use but still represent a potential point of failure eliminated by MSR's. Other safety considerations include a reduction in the overall level of contaminated material within the power plant. The central heat exchange process can be conducted without any radiation leaching into the coolant water [5].



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Why They Could Be the Solution to the Climate Crisis:

In order to surmount the modern climate crisis, world nations must overhaul energy sectors entrenched in fossil fuels. Experts today are in agreement that the conventional renewable technologies of wind and solar energy will not be sufficient base-line energy sources capable of occupying the entire energy grid. 70% of today's leading climate scientists believe that nuclear energy must undergo wide scale implementation if carbon emissions are to see any appreciable decrease [8]. Both wind and solar energy do not possess the dependable, non-weather variable capabilities that require fossil fuels to be so integral in energy supply. Nuclear power addresses these concerns, making it the single greatest, true carbon free option to take the place of fossil fuels on the grid. MSR's represent the pinnacle of nuclear potential and are a system that directly addresses the shortcomings of current reactor designs. It is essential at this point in the global climate crisis that society looks to reassess outdated stigmas and look to these emerging nuclear technologies as the future of renewable energy.

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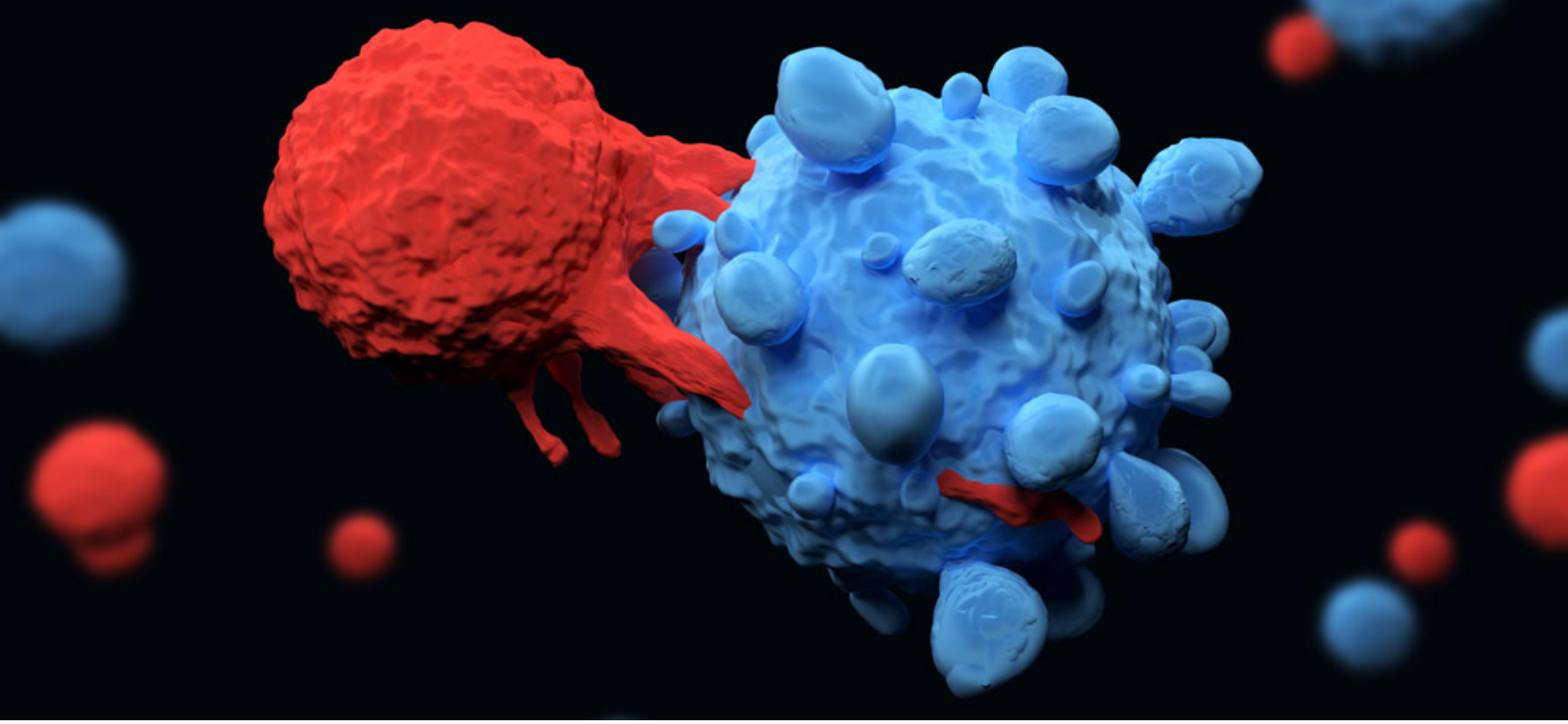


9. CAR-T Cell Therapy: A Potential Cure to Cancer

Rhea Choudhury, ‘23

For a long time, cancer treatment has consisted of surgery, chemotherapy, and radiation therapy. Different types of targeted therapies and drugs have also been standard treatments. Not all are completely effective for long periods of time. However, immunotherapy, a type of treatment that boosts the effectiveness of a patient’s immune system and enables it to better get rid of cancerous cells, has shown promising results. Specifically, CAR-T cell therapy, a type of immunotherapy, has stood out due to its long-term effectiveness in the remission of blood cancer. Scientists have even acknowledged the results as a functioning cure in some patients, bringing us one step closer to saving the lives of millions.

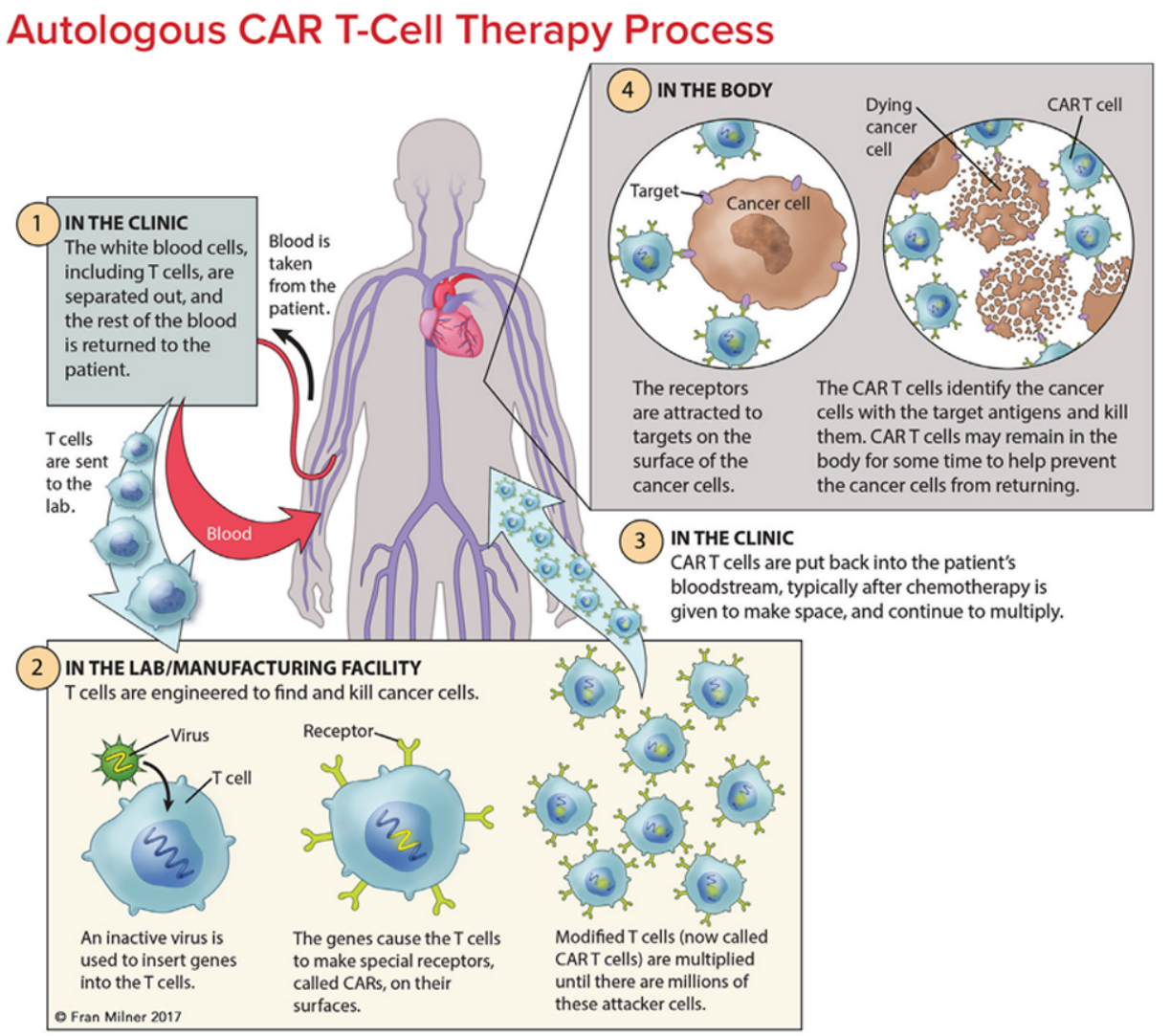
What is CAR-T Cell Therapy?

Immunotherapy helps the immune system protect the body against cancer. Normally, the immune system detects and destroys cancerous cells and stops them from spreading. However, cancer cells can occasionally harm the immune system by affecting how immune cells interact with them. For example, cancer cells can contain genes that make them less visible to immune cells or proteins that disable them. Immunotherapy seeks to stop cancer cells from avoiding immune cells by enhancing the immune system’s capabilities. This can involve removing checkpoints and boundaries that limit immune activity or transferring new cells and antibodies to the patient’s body.

Recent developments in immunotherapy have given rise to a type of treatment called Adoptive Cell Transfer (ACT), in which a patient's own immune cells are used to treat their cancer. Chimeric Antigen Receptor (CAR) T-cell therapy is a type of ACT which, while being mostly limited to small clinical trials, has produced incredible results in some patients. This therapy uses specialized receptors, called CARs, which are added to T cells and designed to bind certain proteins on cancer cells [4]. This fact has garnered CAR-T the attention of many in the medical community; Renier J. Brentjens of Memorial Sloan Kettering Cancer Center in New York, a researcher of CAR-T cell therapy, describes the unique treatment as akin to “giving patients a living drug” [1].

How does CAR-T Cell Therapy Work?

Several steps are involved in carrying out a CAR-T treatment. First, the patient's blood is taken, and their T cells are separated and collected in a procedure called apheresis. Next, the collected cells are genetically engineered in a laboratory. New DNA is introduced to the cells, creating CARs on their surfaces. More of these cells are grown in the lab, and when there are enough for a patient, the cells are frozen and sent to treatment centers. Eventually, they are thawed and injected into the patient [4].



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This form of treatment has many potential side effects, and like most cancer treatments, some can be life-threatening. One of the most common is cytokine release syndrome (CRS). CAR-T cells trigger the production of cytokines, messengers that trigger the immune system. Sometimes, the treatment can cause massive amounts of cytokines to be released into the bloodstream. This can lead to nausea and fevers, and in more serious cases be associated with cardiac failure and poor lung oxygenation [1]. Another side effect is neurological problems, such as trouble speaking and balance problems, due to CAR-T cell therapy's effect on the nervous system. The details of this are relatively unknown and vary between patients [2].

CAR-T Cell Therapy Trials

In 2010, two leukemia patients received CAR-T cell therapy, and their cancers went into remission. After an examination over ten years later, the genetically engineered immune cells were still effective[6]. The study was published in February of 2022.

According to Carl June of the University of Pennsylvania, immunologist and study coauthor, the results were enough to "conclude that CAR-T cells can actually cure patients with leukemia"[5]. David Porter, also from the University of Pennsylvania and the study's other coauthor, said that cancer doctors "don't use words like 'cure' lightly." He also added that for the patients to live cancer-free for over a decade was "beyond [their] wildest expectations"[5].

While CAR-T is typically a last resort for cancer patients who have been unsuccessful with other treatment options, its effectiveness has great implications for the future of cancer treatment.

The Future of CAR-T Cell Therapy

In terms of optimization and adjustments, CAR-T has plenty in store for the future. The treatment method is not as effective outside of leukemia and myeloma, so scientists are working to develop CAR-T that functions for different types of cancer[3]. This could involve genetically engineering cells that are more advanced or altering parts of the T cell to let it function in different environments. In addition, the treatment is very expensive and can have dangerous side effects. Scientists are also working on potential solutions to these problems.

Researchers like Marcela Maus, an immunologist and cell therapist at Mass General Cancer Center, have expressed concern that CAR is uncontrollable. After the CAR-T cells are let out into the body, they're like "teenage kids," said Maus[3]. This could present problems if the cells function incorrectly. Maus and others want to be able to control when the CAR attacks cancerous cells. Similarly, Peter Yingxiao Wang, a synthetic biologist at the University of California in San Diego, agrees that there is still much work to be done, calling it an "infant field" of research[3]. Even though CAR-T cell therapy has shown promising results, there are still adjustments and possible advancements that can increase its success rate, reliability, and make it more readily available to the public, giving many cancer patients a shot at recovery.

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10. Ice Needles, Patterned Soil, and the Red Planet

Sharmila Green, ‘23

The earth’s surface contains a wide range of geomorphic features. From the tallest mountains to the lowest crater, processes of erosion and transport shape our planet to the extremes. Topsoil, the uppermost layer of the earth, is prone to many changes. Some occur overnight or over otherwise short periods, like animals’ footprints or plants rooting themselves. Other patterns found in the topsoil are due to long-term processes, like faulting or running water. However, while patterns like the shape of a valley may be obvious due to a river carving out its path, some formations may be more obscure shapes, like polygons or circles. Scientists around the world, from Seattle to China to the Netherlands, are proposing “ice needles”–small columns of ice formed by groundwater–as the reason for certain soil formations. Interestingly, this proposition may also provide more evidence for the existence of water on our neighboring planet, Mars.

Patterns of Soil

In Arctic climates, scientists can study intricate patterns in the soil. Since wind and other forces are often lacking,Zand temperatures change rapidly, this type of environment is very intriguing [2].

Soil patterns in the Arctic vary widely, showcasing winding labyrinths, polygon shapes, and circular formations, among others. Scientists think this may be due to freeze-thaw cycles seen in these cold environments which form ice needles [1]. These ice needles are not very uncommon–Scientist and report coauthor Bernard Hallet says they are “very striking” and “quite common,” often found in “crunchy” ground

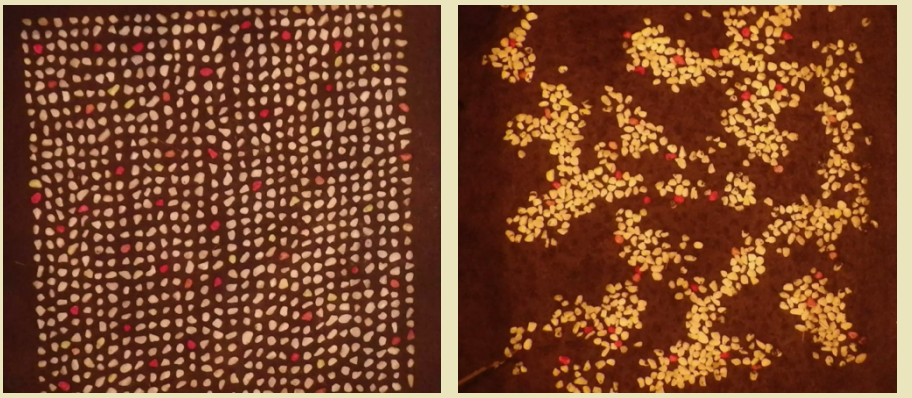
Circled soil and stone from Svalbard, Norway [2]

early in the morning [2]. These freeze-thaw cycles occur when air temperatures drops below freezing, therefore allowing the water in the soil to freeze. Because water expands as it freezes, this can cause movement in the soil. These cycles occur yearly, with freezing in winter and thawing in summer, as well as daily, because of changes in temperature from night to daytime [3]. Scientists replicated these cycles within the laboratory to see how soil shifts specifically due to these patterns.

The Ice Needle Experiment

In the Shaoxing, China lab, researchers Anyuan Li, Norikazu Matsuoka, and Fujun Niu used both level and sloping ground, simulating patterns with concentration of stone ranging from 20 to 80%. In a controlled cold room, they replicated needles of ice, and recorded the height of needles, soil temperature, and moisture, while taking pictures of the patterned rock in ten-minute intervals [1].

Rocks being shifted by ice needles at their highest point in the freeze-thaw cycle [1]



Laboratory rock pattern of 30% stone cover at level ground before and after all 10-minute intervals of freeze-thaw cycles using ice needles [1]

The team mapped the displacement of each stone, analyzing the relationship between clusters of stones and the speed at which they moved to their new locations. They found that stones moved faster when not concentrated alongside others. They also noticed interesting patterns with the ice needle growth. As stone concentration increased from 10% to 80%, the needles decreased in height and straightened, meaning clusters of stones were less likely to move due to the ice needles, making those features more stable after formation [1].

The scientists noted that the physical processes attributed to the sloping ground–such as sliding or "creep" of rocks–also helped with movement, but the ice needles remained the most crucial aspect of their organization, helping them form into unique patterns.

Earth and Beyond

In the end, the team concluded that because of the Arctic's dependence on freeze-thaw cycles, changes in rock pattern formations are indicative of changes in these conditions. This is especially important in the discussion of climate change since they note that changes in the rock patterns "may be a crucial first sign of local climate change impacts" [1]. Changes could also hint toward the release of greenhouse gases from permafrost since this frozen land holds many of these patterns and would be affected as Arctic cycles begin to change.

Furthermore, these patterned ground types have been discovered on Mars by the Curiosity rover. Not all atmospheres can form these ice needles–only areas where the ground has water and the air can cool rapidly [2]. While Mars does not necessarily have an atmosphere particularly rich in water, these patterns provide evidence that tiny ice crystals still change the surface of Mars.

In their research, scientists like Li and Hallet have shown how ice needles are important parts of geomorphology and our planet's surface. These tiny formations make a big impact, giving information on the climate and weather, especially in regards to freeze-thaw cycles that shape our world and our neighboring red planet's. While there is still more research needed in understanding how geologic surfaces form, these common structures provide important insight into the forces that shape our environment.

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Header Image:

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